Complete expansions of the chart integrals with arbitrary (k, h): $Z(\sqrt{n}) = n^{-p/2} K D \sum_{i \le j} c_{j,i} \left(\frac{1}{2} \log n + S \log b\right)^i + O(n^{-p/2 - \varepsilon/2})$

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Abstract

Units 50 and 52 reduced the chart integrals exactly to the form $Z(\sqrt{n}) = K(\sqrt{n})^{-p} D\, H_j(\sqrt{n}\, b^S)$ for a divide-and-integrate tower H on an admissible base with tail rate ε. This unit turns the base-independent expansion of units 49/51 into a single generic theorem: $Z(\sqrt{n}) = n^{-p/2} K D \sum_{i \le j} c_{j,i} (\frac{1}{2} \log n + S \log b)^i + O(n^{-p/2-\varepsilon/2})$, instantiated for the all-equal case with arbitrary (k, h) $(\varepsilon = 1)$ and the mixed case $(\varepsilon = q - p)$. A bridge lemma identifies unit 39's `iterChart` with `iterChartGen` at $k = 1$, $h = 0$. Zero `sorry`/`axiom`.

1 Setting

With $L = \sqrt{n}\, b^S$ one has $\log L = \frac{1}{2} \log n + S \log b$ and $L^{-\varepsilon} = b^{-S\varepsilon} n^{-\varepsilon/2}$; the expansion $|H_j(L) - \sum c_{j,i} \log^i L| \le (M/\varepsilon^j) L^{-\varepsilon}$ therefore transfers to an explicit $O(n^{-p/2-\varepsilon/2})$ remainder once multiplied by the prefactor $K(\sqrt{n})^{-p} D$.

2 The things formalised

```
noncomputable def towerCoeff (G :   →  ) (A :  ) (j :  ) (b :  ) (S :  ) (n :  ) :   :=
    i   Finset.range (j + 1),
    divCoeff G A j i * (Real.log n / 2 + (S :  ) * Real.log b) ^ i
theorem tower_expansion_isBigO (hG : LogBase G A M C   ) (j :  ) (K D p b :  ) (S :  )
    (hb : 0 < b) (Z :   →  )
    (hZ :   n, 0 < n →
      Z n = K * (Real.sqrt n) ^ (-p) * D * iterDivPrim G j (Real.sqrt n * b ^ S)) :
    (fun n => Z n - n ^ (-(p / 2)) * (K * D * towerCoeff G A j b S n))
      =O[atTop] fun n => n ^ (-(p / 2) -   / 2)
theorem iterChartGen_expansion_isBigO ...        --   = 1, remainder n^(-(p/2) - 1/2)
theorem iterChartGen_mixed_expansion_isBigO ...  --   = q - p
theorem iterChart_eq_iterChartGen (  a b :  ) (d :  ) (c :  ) :
    iterChart   a b d c = iterChartGen   a b (fun _ => 1) (fun _ => 0) d c
```

3 Proof strategy

The generic theorem is `IsBigO.of_bound` with the constant $|KD|\,(M/\varepsilon^j)\, b^{-S\varepsilon}$, valid for $n \ge \max(1, b^{-2S})$ so that $L \ge 1$; the three `rpow` identities $((\sqrt{n})^{-p} = n^{-p/2}$, $(\sqrt{n})^{-\varepsilon} = n^{-\varepsilon/2}$, $(b^S)^{-\varepsilon} = b^{-S\varepsilon})$ are isolated as `haves`. The two instances supply only the exact-reduction hypothesis `hZ` from units 50 and 52. The bridge lemma is an induction with `pow_zero`, `pow_one`.

4 Roadblocks and resolutions

- A `ring` after a `rw` that had already closed the goal (`log_pow` produced the exact form); remove it. Otherwise first-pass.

5 What was Mathlib, what was new

Mathlib: `Real.rpow_add`, `Real.rpow_mul`, `Real.sqrt_eq_rpow`, `Real.log_pow`. New: the generic tower expansion and its two instances.

6 Lessons

Once the exact reductions are stated in a common shape, the asymptotic and expansion theorems become one generic lemma plus one-line instances; stating the reductions with that shape in mind (constants K, D, scale b^S) is worth the small awkwardness at the reduction stage.

7 Outcome

For arbitrary (k, h) the chart integrals of `thm:TaylorTree` in the all-equal and one-noncritical regimes now have complete polynomial-in-$\log n$ expansions with explicit recursive coefficients and power-saving remainders. Remaining at this level: several noncritical coordinates, and the product-set (Fubini) form of the d-fold integral.